

# gdTAI V4 Step 1 Preflight and Split Freeze

- gdTAI V4 Step 1 Preflight and Split Freeze
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# gdTAI V4 Step 1 Preflight and Split Freeze

**Overall status:** PASS  
**Protocol:** v1.2  
**Training started:** No  
**Step 2 authorized:** No; supervision approval is still required

## Decision

This workflow performed only the precommitted read-only input, truth-label, feature, exclusion-feasibility, and grouped-split checks. It did not fit, calibrate, threshold, compare, or promote a classifier.

Failures block Step 2. Warnings require explicit supervision. The frozen CD4/Treg thresholds were evaluated exactly as written and were not tuned.

### Blocking failures

No rows.

### Warnings

No rows.

## All preflight checks

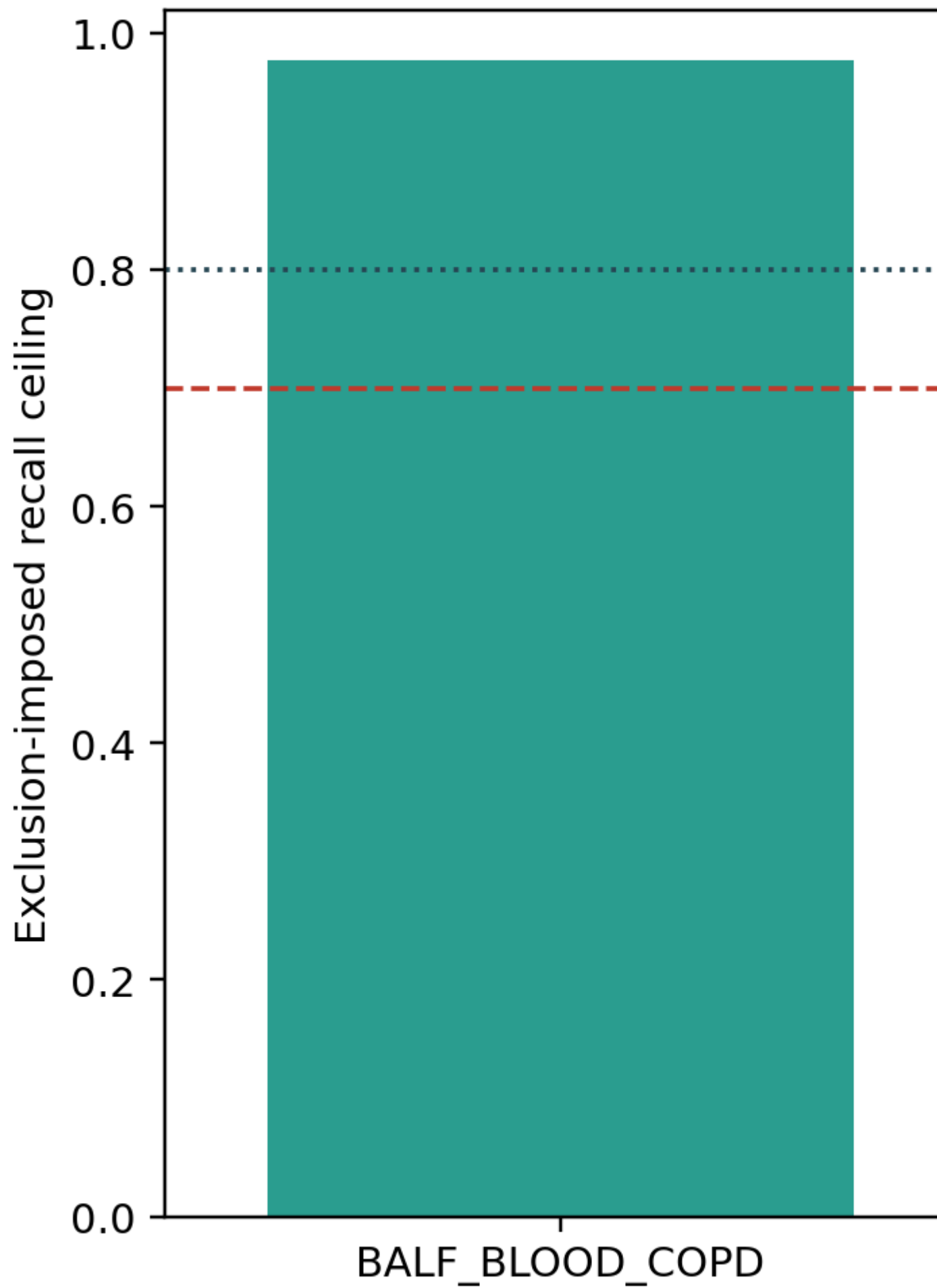
| check_id                  | status | observed | required                     | details                         |
|---------------------------|--------|----------|------------------------------|---------------------------------|
| protocol_version          | PASS   | 1.2      | 1.2                          | Frozen protocol version         |
| feature_count             | PASS   | 197      | 197                          | Frozen individual-gene universe |
| training_feature_coverage | PASS   |          | >=0.9 and all critical genes | Hard-gated development inputs   |

| check_id                           | status | observed                                                                                                                                                             | required                                                                           | details                                              |
|------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|
| expression_input_contract          | PASS   | ['hra005041']                                                                                                                                                        | raw counts, except registered HRA005041 passing full log1p(CP10K) audit            | Full sparse data-array and inverse-library-size scan |
| registered_hashes                  | PASS   |                                                                                                                                                                      | all available expected hashes match                                                | Full SHA-256 unless explicitly skipped               |
| input_file_state                   | PASS   | 16                                                                                                                                                                   | 16                                                                                 | Size and mtime unchanged after read-only workflow    |
| gse144469_join                     | PASS   | 107068                                                                                                                                                               | 107068 unique SRR + barcode mappings                                               | Raw expression to canonical metadata                 |
| primary_labels_nonempty            | PASS   | {'abT_primary': {'BALF_BLOOD_COPD': 32404, 'GSE144469': 60175, 'HRA005041': 543028}, 'gdT_primary': {'BALF_BLOOD_COPD': 1033, 'GSE144469': 4003, 'HRA005041': 4894}} | both classes in each primary source                                                | Expression-independent productive TCR rules          |
| primary_label_overlap              | PASS   | 0                                                                                                                                                                    | 0                                                                                  | Mutually exclusive primary labels                    |
| sensitivity_excluded_from_training | PASS   | 0                                                                                                                                                                    | 0                                                                                  | Silver and all sorted cohorts are sensitivity-only   |
| cd4_treg_recall_ceiling            | PASS   | 0.983805                                                                                                                                                             | macro >= 0.8; each source >= 0.7                                                   | Fixed post-model exclusions before Step 2            |
| group_leakage                      | PASS   | 1                                                                                                                                                                    | 1                                                                                  | No group appears in multiple inner folds             |
| outer_fold_nonempty                | PASS   | 1                                                                                                                                                                    | True                                                                               | Every stage has train and held-out cells             |
| nk_stratum_composition             | PASS   | {'representative': 0.6, 'gdt_like_TRDCpos_TRDV_neg': 0.2, 't_like_hard': 0.2}                                                                                        | {'representative_fraction': 0.6, 't_like_fraction': 0.2, 'gdt_like_fraction': 0.2} | All strict NK negatives assigned once                |
| supplemental_nk_pool               | PASS   | 323771                                                                                                                                                               | >0                                                                                 | Mapped, no-productive-TCR atlas NK negatives         |
| legacy_v4_archived                 | PASS   | True                                                                                                                                                                 | True                                                                               | No overwrite of the earlier experimental artifact    |

## CD4/Treg recall cost

The recall ceiling is  $1 - \text{union\_excluded} / \text{gdT\_primary}$ . It is the maximum post-exclusion recall even for a perfect pre-exclusion classifier.

Fixed CD4/Treg

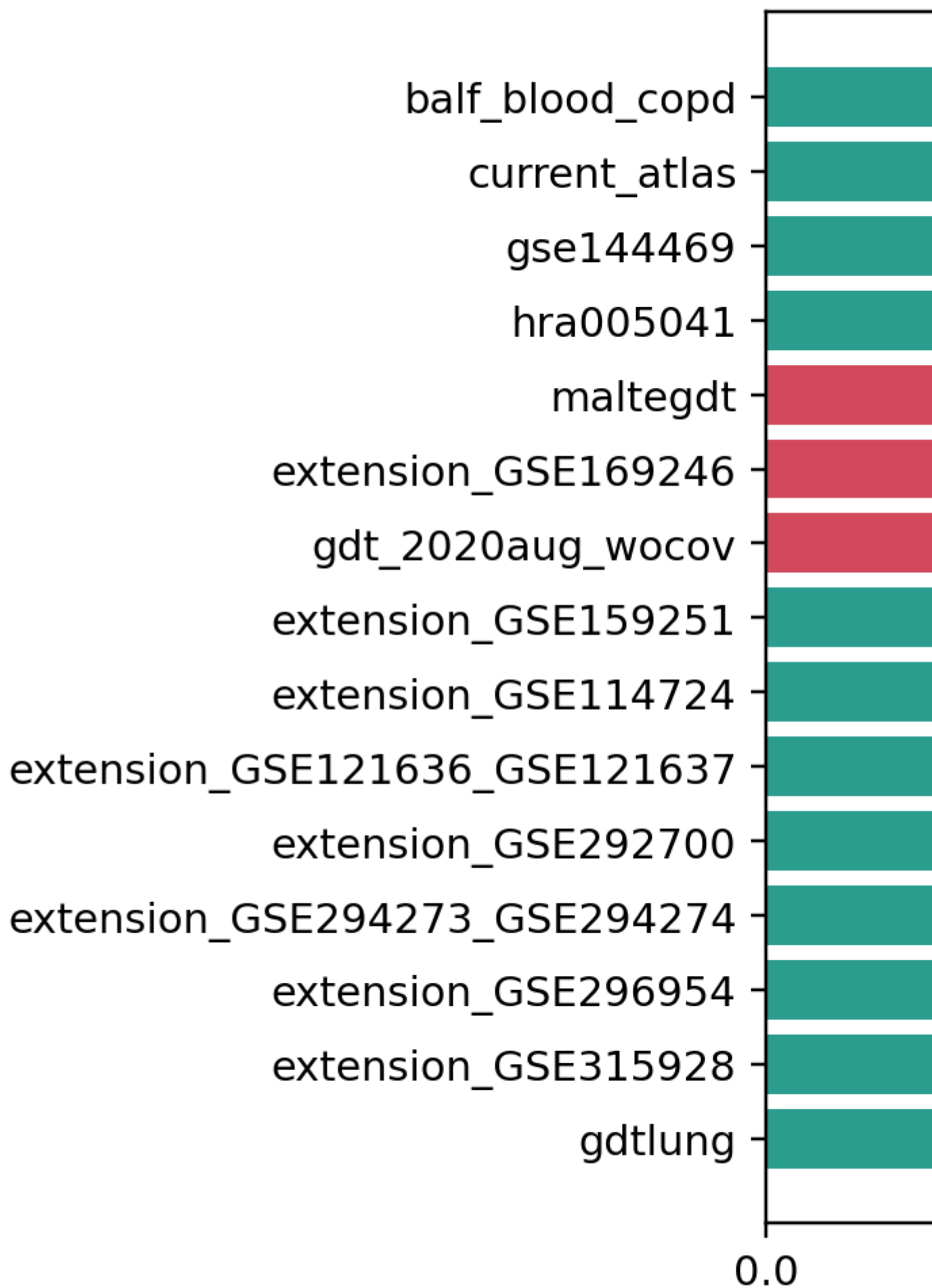


## Recall ceiling

| source_gse_id   | n_cells | cd4_helper_only | treg_only | overlap | union_excluded | recall_ceiling | applicable_floor | margin_above_floor |
|-----------------|---------|-----------------|-----------|---------|----------------|----------------|------------------|--------------------|
| BALF_BLOOD_COPD | 1033    | 22              | 1         | 0       | 23             | 0.9777         | 0.7000           | 0.2777             |
| GSE144469       | 4003    | 65              | 6         | 18      | 89             | 0.9778         | 0.7000           | 0.2778             |
| HRA005041       | 4894    | 20              | 0         | 0       | 20             | 0.9959         | 0.7000           | 0.2959             |
| SOURCE_MACRO    | 9930    | 107             | 7         | 18      | 132            | 0.9838         | 0.8000           | 0.1838             |

# Frozen feature coverage

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## Feature coverage

| input_id                      | role                                            | features_present | features_expected | feature_coverage | missing_critical_count | hard_feature_gate |
|-------------------------------|-------------------------------------------------|------------------|-------------------|------------------|------------------------|-------------------|
| current_atlas                 | stage1_NK_expression_and_future_atlas_inference | 197              | 197               | 1.0000           | 0                      | True              |
| hra005041                     | primary_development                             | 197              | 197               | 1.0000           | 0                      | True              |
| gse144469                     | primary_development                             | 197              | 197               | 1.0000           | 0                      | True              |
| half_blood_copd               | primary_development_reused_benchmark            | 197              | 197               | 1.0000           | 0                      | True              |
| gdt_2020aug_wocov             | sorted_sensitivity_only                         | 163              | 197               | 0.8274           | 0                      | False             |
| maltegdt                      | sorted_sensitivity_only                         | 82               | 197               | 0.4162           | 0                      | False             |
| gdtlung                       | sorted_sensitivity_only                         | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE114724           | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE121636_GSE121637 | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE159251           | frozen_negative_stress                          | 188              | 197               | 0.9543           | 0                      | False             |
| extension_GSE169246           | reduced_feature_schema_sensitivity              | 145              | 197               | 0.7360           | 0                      | False             |
| extension_GSE292700           | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE294273_GSE294274 | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE296954           | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |
| extension_GSE315928           | frozen_negative_stress                          | 197              | 197               | 1.0000           | 0                      | False             |

## Expression input audit

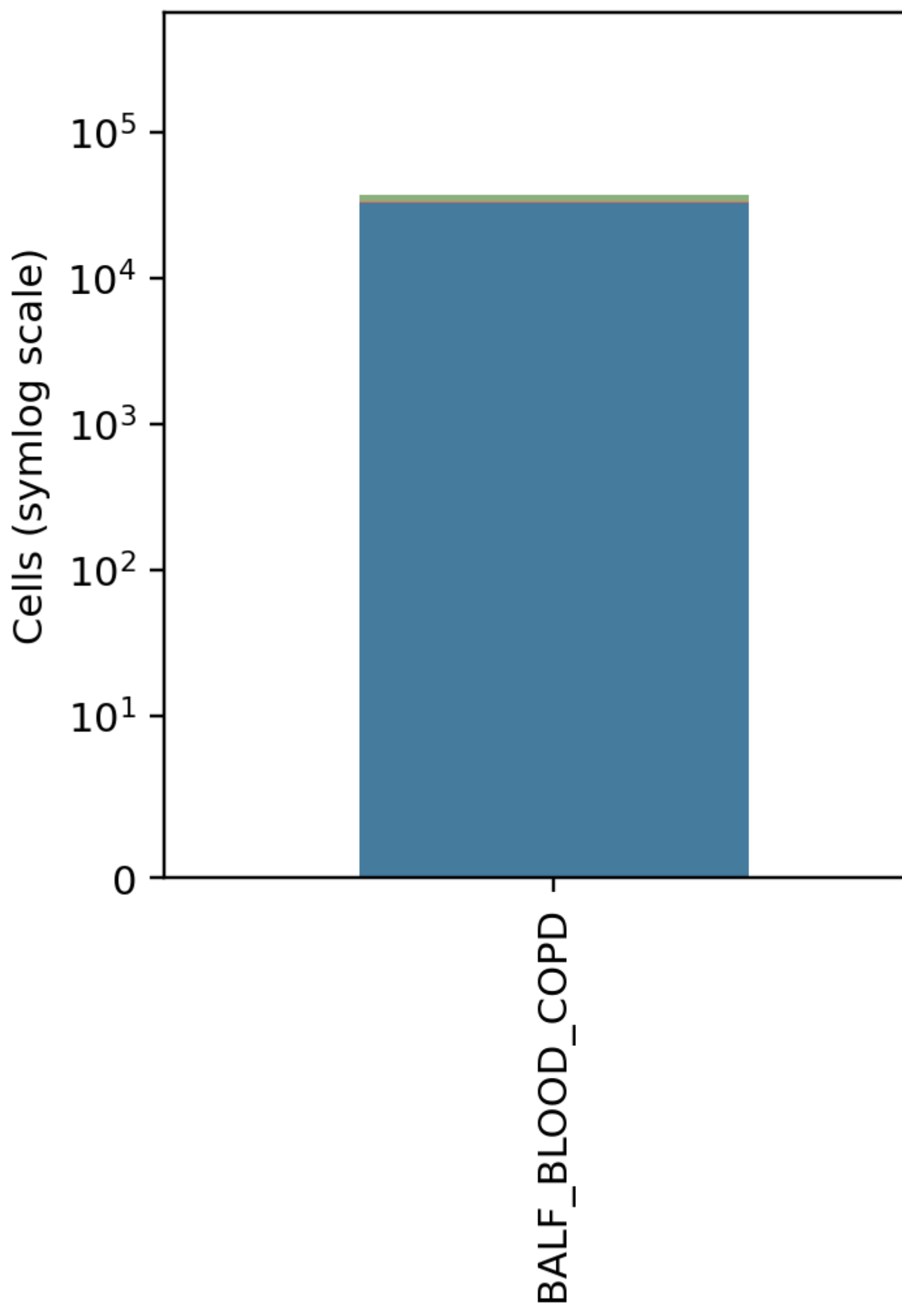
Every stored sparse value was scanned. Raw inputs must be finite, nonnegative, and integer-like within `1e-6`.

The registered HRA005041 exception must additionally reconstruct a per-cell library sum of 10,000 from `expm1(X)` within the frozen absolute tolerance.

| input_id                      | matrix_key    | configured_matrix_state | n_obs   | nnz        | raw_count_pass | transformed_max_abs_deviation | transformed_rows_outside_tolerance | expression_contract_pass |
|-------------------------------|---------------|-------------------------|---------|------------|----------------|-------------------------------|------------------------------------|--------------------------|
| current_atlas                 | X             | raw_counts              | 3705306 | 5503996785 | True           |                               | 0                                  | True                     |
| hra005041                     | X             | log1p_cp10k_registered  | 766639  | 1442826937 | False          | 0.0000                        | 0                                  | True                     |
| gse144469                     | layers/counts | raw_counts              | 107068  | 215368848  | True           |                               | 0                                  | True                     |
| half_blood_copd               | layers/counts | raw_counts              | 46273   | 125599864  | True           |                               | 0                                  | True                     |
| gdt_2020aug_wocov             | X             | raw_counts              | 25904   | 39050646   | True           |                               | 0                                  | True                     |
| maltegmt                      | X             | raw_counts              | 7800    | 12020698   | True           |                               | 0                                  | True                     |
| gdtlung                       | X             | raw_counts              | 15175   | 15468404   | True           |                               | 0                                  | True                     |
| extension_GSE114724           | X             | raw_counts              | 28652   | 50878640   | True           |                               | 0                                  | True                     |
| extension_GSE121636_GSE121637 | X             | raw_counts              | 18793   | 28893248   | True           |                               | 0                                  | True                     |
| extension_GSE159251           | X             | raw_counts              | 68898   | 81434083   | True           |                               | 0                                  | True                     |
| extension_GSE169246           | X             | raw_counts              | 354373  | 430559363  | True           |                               | 0                                  | True                     |
| extension_GSE292700           | X             | raw_counts              | 78254   | 127198003  | True           |                               | 0                                  | True                     |
| extension_GSE294273_GSE294274 | X             | raw_counts              | 55744   | 111603853  | True           |                               | 0                                  | True                     |
| extension_GSE296954           | X             | raw_counts              | 86608   | 122094020  | True           |                               | 0                                  | True                     |
| extension_GSE315928           | X             | raw_counts              | 66813   | 91568523   | True           |                               | 0                                  | True                     |







## Label composition

| source_gse_id      | truth_class        | n_cells |
|--------------------|--------------------|---------|
| BALF_BLOOD_COPD    | abT_primary        | 32404   |
| BALF_BLOOD_COPD    | dual_or_ambiguous  | 2938    |
| BALF_BLOOD_COPD    | gdT_primary        | 1033    |
| BALF_BLOOD_COPD    | gdT_silver         | 31      |
| BALF_BLOOD_COPD    | single_abT_weak    | 3923    |
| BALF_BLOOD_COPD    | unlabeled          | 280     |
| GDT_2020AUG_woCOV  | sorted_sensitivity | 25904   |
| GDTlung2023july_7p | sorted_sensitivity | 15175   |
| GSE125527          | unlabeled          | 3797    |
| GSE144469          | abT_primary        | 60175   |
| GSE144469          | dual_or_ambiguous  | 5857    |
| GSE144469          | gdT_primary        | 4003    |
| GSE144469          | gdT_silver         | 415     |
| GSE144469          | single_abT_weak    | 3379    |
| GSE144469          | unlabeled          | 8662    |
| GSE145926          | unlabeled          | 361     |
| GSE162498          | unlabeled          | 129     |
| GSE171037          | unlabeled          | 225     |
| GSE178882          | unlabeled          | 5493    |
| GSE188620          | unlabeled          | 20162   |
| GSE190870          | unlabeled          | 1       |
| GSE211504          | unlabeled          | 303     |
| GSE212217          | unlabeled          | 25627   |
| GSE228597          | unlabeled          | 17507   |
| GSE234069          | unlabeled          | 136     |
| GSE235863          | unlabeled          | 18132   |
| GSE240865          | unlabeled          | 364     |
| GSE243013          | unlabeled          | 42490   |
| GSE243572          | unlabeled          | 17075   |
| GSE243905          | unlabeled          | 18659   |

## GSE144469 join

| source_gse_id | source_cells | source_unique_keys | canonical_cells | canonical_unique_keys | mapped_cells | unmapped_cells | mapping_fraction | source_duplicate_keys | canonical_duplicate_keys | obs_sample_disagrees_with_id_srr | obs_sample_barcode_unique_keys |
|---------------|--------------|--------------------|-----------------|-----------------------|--------------|----------------|------------------|-----------------------|--------------------------|----------------------------------|--------------------------------|
| GSE144469     | 107068       | 107068             | 107068          | 107068                | 107068       | 0              | 1.0000           | 0                     | 0                        | 16851                            | 90217                          |

## Frozen outer folds

| outer_fold_id           | heldout_source  | stage  | outer_train_cells | outer_train_positive | outer_eval_cells | outer_eval_positive | outer_train_groups |
|-------------------------|-----------------|--------|-------------------|----------------------|------------------|---------------------|--------------------|
| outer_0_HRA005041       | HRA005041       | stage1 | 437630            | 105801               | 634293           | 629342              | 741                |
| outer_0_HRA005041       | HRA005041       | stage2 | 104917            | 5036                 | 627813           | 4894                | 28                 |
| outer_1_GSE144469       | GSE144469       | stage1 | 995704            | 666728               | 76219            | 68415               | 722                |
| outer_1_GSE144469       | GSE144469       | stage2 | 665173            | 5927                 | 67557            | 4003                | 25                 |
| outer_2_BALF_BLOOD_COPD | BALF_BLOOD_COPD | stage1 | 1034283           | 697757               | 37640            | 37386               | 754                |
| outer_2_BALF_BLOOD_COPD | BALF_BLOOD_COPD | stage2 | 695370            | 8897                 | 37360            | 1033                | 41                 |

## Input identity

| input_id                  | role                                            | size_bytes  | sha256           | expected_hash_match |
|---------------------------|-------------------------------------------------|-------------|------------------|---------------------|
| current_atlas             | stage1_NK_expression_and_future_atlas_inference | 73350311676 | 9060920d6fe19bdb | True                |
| legacy_metadata_reference | read_only_TCR_and_GSE144469_metadata_reference  | 24383355208 | f44bd4941a7b8c8d | True                |
| hra005041                 | primary_development                             | 19626113528 | fc9ead68c3d85232 | True                |
| gse144469                 | primary_development                             | 1536829723  | 456b2da064c9ad60 | True                |
| balf_blood_copd           | primary_development_reused_benchmark            | 2110825599  | acf33fcc3851f77c | True                |
| gdt_2020aug_wocov         | sorted_sensitivity_only                         | 591495928   | 650f583b85f05ede | True                |
| maltegdt                  | sorted_sensitivity_only                         | 179862512   | bbd460ef153ce6b2 | True                |
| gdtlung                   | sorted_sensitivity_only                         | 276869875   | d5915e6b50241ff8 | True                |
| extension_GSE114724       | frozen_negative_stress                          | 131909269   | 394425c7937432ac | True                |

| input_id                      | role                               | size_bytes | sha256           | expected_hash_match |
|-------------------------------|------------------------------------|------------|------------------|---------------------|
| extension_GSE121636_GSE121637 | frozen_negative_stress             | 77281297   | b7182274cab834f1 | True                |
| extension_GSE159251           | frozen_negative_stress             | 223142214  | abe6502b3a15a3ba | True                |
| extension_GSE169246           | reduced_feature_schema_sensitivity | 2176683186 | f251382c62109a48 | True                |
| extension_GSE292700           | frozen_negative_stress             | 326784213  | 78c9ada980a21592 | True                |
| extension_GSE294273_GSE294274 | frozen_negative_stress             | 289243072  | 6a20d61544c7cd33 | True                |
| extension_GSE296954           | frozen_negative_stress             | 304970784  | f3484482e35b49d3 | True                |
| extension_GSE315928           | frozen_negative_stress             | 253771849  | 2fe674761db67aa3 | True                |

## Canonical artifacts

- Cell-label manifest: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/cell_label_manifest.csv.gz`
- Inner group assignments: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/inner_group_split_manifest.csv.gz`
- Full checks: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_preflight/preflight_checks.csv`
- Machine summary: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_preflight/preflight_summary.json`
- Reader HTML: `gdT_prediction/gdtai_v4_preflight/index.html`
- PDF: `gdT_prediction/gdtai_v4_preflight/gdtai_v4_preflight_report.pdf`

## Gate

Stop before Step 2. No model fitting is permitted until the user reviews this package and explicitly approves the next gate.